

This appendix provides supplementary material supporting the empirical analysis in Section 5.

- Appendix A provides additional background on why credit classification is a suitable application domain for evolutionary feature construction and supports the transferability discussion in Section 5.5.
- Appendix B presents supplementary analysis of importance-value power transformations and effective protection levels, and supports the methodological comparison in Section 5.3.
- Appendix C compares the deterministic top- K and probabilistic APM variants across different base learners.

A Feature Construction in Credit Classification

Credit classification is a suitable application for evolutionary feature construction because credit risk often depends on nonlinear interactions among attributes, and useful predictors may arise from complex combinations of raw features that are labor-intensive to design manually. Prior work shows that engineered transaction and behavioral features improve cost-sensitive fraud detection [3] and cash-out fraud ranking [32], while feature engineering with virtual sample generation improves robustness in data-scarce credit scoring [33]. A two-stage pipeline combining XGBoost-based feature construction with graph-based deep models also outperforms models trained on the original features [17]. However, these studies largely lack iterative refinement during search, motivating an evolutionary framework with adaptive protection to preserve high-value features while continuing the search for better feature combinations.

B Further Analysis of Power Transformation and Protection Level

B.1 Power Transformation

As a complementary investigation within RQ3, we examine whether nonlinear transformations of importance values improve the protection mechanism by applying a power transformation $|\beta_j|^p$ for $p \in \{0.5, 1, 2, 5\}$. Figure 6 and Table 8 show that power 1, corresponding to linear scaling, provides the best overall performance. Higher powers overemphasize differences, creating extreme rankings that underweight features of medium importance, leading to inadequate protection and performance degradation. Power 0.5 also underperforms relative to power 1, as it compresses importance values toward uniformity, reducing the contrast between high- and low-importance features and weakening the protection signal. Linear scaling therefore provides the most balanced approach, confirming that raw coefficient magnitudes are appropriate without requiring nonlinear transformations.

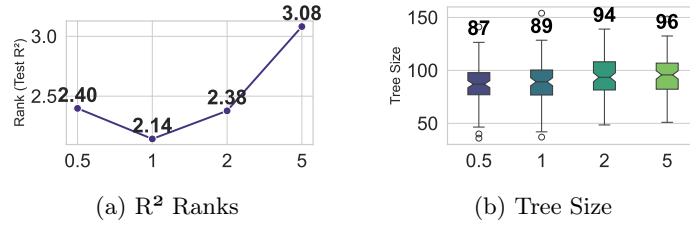


Fig. 6: Comparison of feature importance power values.

Table 8: Statistical comparison of feature importance power values.

	1	2	5
0.5	3(+)/92(~)/3(-)	4(+)/91(~)/3(-)	15(+)/81(~)/2(-)
1	—	1(+)/97(~)/0(-)	27(+)/70(~)/1(-)
2	—	—	20(+)/75(~)/3(-)

B.2 Effective Protection Level

To determine an effective restoration intensity, we examine the trade-off between preserving existing valuable trees and allowing further tree modifications by comparing different protection coefficients. We consider protection coefficients of 0, 1, 3, and 5, where 0 represents no protection. Figure 7 and Table 9 summarize the outcomes. Statistical comparisons show that protection consistently outperforms no protection: with 10 trees per individual, larger protection coefficients perform better. Coefficients of 1 or 3 provide insufficient protection, whereas a coefficient of 5 provides adequate protection while still allowing sufficient modification of other trees.

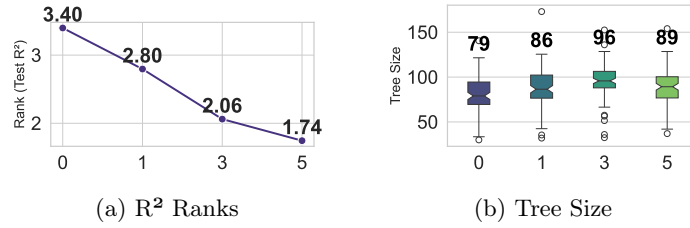


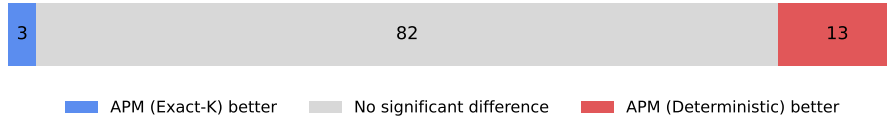
Fig. 7: Comparison of different protection coefficients.

Table 9: Statistical comparison of different protection coefficients.

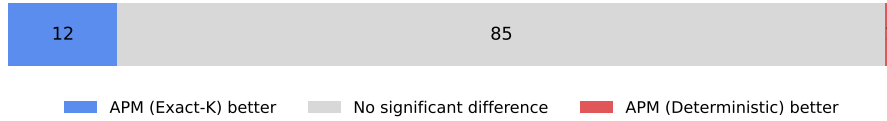
	1	3	5
0	0(+)/67(~)/31(-)	2(+)/40(~)/56(-)	2(+)/38(~)/58(-)
1	—	2(+)/53(~)/43(-)	3(+)/39(~)/56(-)
3	—	—	0(+)/77(~)/21(-)

C Deterministic vs. Probabilistic Adaptive Protection

We compare the probabilistic APM against a deterministic top- K variant. Both follow a partial elitism idea that keeps exactly $K = 5$ features per individual unchanged through the genetic operations: the deterministic variant always preserves the 5 highest-importance features, whereas the probabilistic APM samples 5 features without replacement, with each feature drawn with probability proportional to its importance. In both variants, features with zero importance are never preserved. As shown in Figure 8, the two variants are broadly equivalent, with the large majority of datasets showing no significant difference under either base learner. The minor differences that do arise vary by base learner, suggesting these stem from how feature importance is estimated under each base learner rather than a systematic advantage of either formulation. Overall, both variants perform similarly, confirming that feature-importance-guided protection is the key mechanism regardless of whether the protected features are selected deterministically or sampled probabilistically.



(a) Ridge regression base learner.



(b) Random decision tree base learner.

Fig. 8: Comparison of deterministic and probabilistic APM variants across base learners.